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DBMS LAB 1-DML COMMANDS

1. Student table:

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100)  
);  
SHOW TABLES;  
INSERT INTO Student (student_id, student_name, email) VALUES  
(1, 'Alice Smith', 'alice@email.com'),  
(2, 'Bob Jones', 'bob@email.com'),  
(3, 'Charlie Brown', 'charlie@email.com');  
UPDATE Student SET email = 'bob.new@email.com' WHERE student_id = 2;  
DELETE FROM Student WHERE student_id = 1;  
SELECT * FROM Student;
```

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| student                |  
+-----+  
  
+-----+-----+-----+  
| student_id | student_name | email                |  
+-----+-----+-----+  
|           2 | Bob Jones    | bob.new@email.com   |  
|           3 | Charlie Brown | charlie@email.com    |  
+-----+-----+-----+
```

2.Course table:

```
CREATE TABLE Course (  
    course_code VARCHAR(10) PRIMARY KEY,  
    title VARCHAR(50) NOT NULL,  
    credits INT  
);  
SHOW TABLES;  
INSERT INTO Course (course_code, title, credits) VALUES  
('CS101', 'Introduction to Computer Science', 4),  
('MATH201', 'Calculus I', 3),  
('ENG102', 'Literature and Composition', 3);  
UPDATE Course SET credits = 5 WHERE course_code = 'CS101'  
DELETE FROM Course WHERE course_code = 'ENG102';
```

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| course |  
+-----+  
  
+-----+-----+-----+  
| course_code | title | credits |  
+-----+-----+-----+  
| CS101 | Introduction to Computer Science | 5 |  
| MATH201 | Calculus I | 3 |  
+-----+-----+-----+
```

3.Professor table:

```

CREATE TABLE Professor (
    professor_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    contact_email VARCHAR(100)
);
SHOW TABLES;
INSERT INTO Professor (professor_id, first_name, last_name, contact_email) VALUES
(101, 'Alan', 'Turing', 'alan@uni.edu'),
(102, 'Marie', 'Curie', 'marie@uni.edu'),
(103, 'Albert', 'Einstein', 'albert@uni.edu');
UPDATE Professor SET last_name = 'Turing-Smith' WHERE professor_id = 101;
DELETE FROM Professor WHERE professor_id = 103;
SELECT * FROM Professor;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| professor              |
+-----+

+-----+-----+-----+-----+
| professor_id | first_name | last_name  | contact_email |
+-----+-----+-----+-----+
|           101 | Alan       | Turing-Smith | alan@uni.edu  |
|           102 | Marie      | Curie        | marie@uni.edu |
+-----+-----+-----+-----+

```

4.Department table:

```

CREATE TABLE Department (
    department_id INT PRIMARY KEY,
    department_title VARCHAR(100),
    office_location VARCHAR(50)
);
SHOW TABLES;
INSERT INTO Department (department_id, department_title, office_location) VALUES
(1, 'Computer Science', 'Building A'),
(2, 'Physics', 'Building B'),
(3, 'Mathematics', 'Building C');
UPDATE Department SET office_location = 'Science Wing' WHERE department_id = 2;
DELETE FROM Department WHERE department_id = 3;
SELECT * FROM Department;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| department            |
+-----+
+-----+-----+-----+-----+
| department_id | department_title | office_location |
+-----+-----+-----+-----+
|          1 | Computer Science | Building A      |
|          2 | Physics          | Science Wing    |
+-----+-----+-----+-----+

```

5.Classroom table:

```

CREATE TABLE Classroom (
    room_number VARCHAR(10) PRIMARY KEY,
    building VARCHAR(50),
    capacity INT
);
SHOW TABLES;
INSERT INTO Classroom (room_number, building, capacity) VALUES
('101A', 'Science Hall', 50),
('204B', 'Arts Block', 30),
('301C', 'Main Building', 100);
UPDATE Classroom SET capacity = 60 WHERE room_number = '101A';
DELETE FROM Classroom WHERE room_number = '204B';
SELECT * FROM Classroom;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| classroom              |
+-----+
+-----+-----+-----+
| room_number | building      | capacity |
+-----+-----+-----+
| 101A        | Science Hall  | 60       |
| 301C        | Main Building | 100      |
+-----+-----+-----+

```

6.Library table:

```

CREATE TABLE LibraryBook (
    book_id INT PRIMARY KEY,
    book_title VARCHAR(200) NOT NULL,
    author_name VARCHAR(100)
);
SHOW TABLES;
INSERT INTO LibraryBook (book_id, book_title, author_name) VALUES
(1, 'Database System Concepts', 'Silberschatz'),
(2, 'Clean Code', 'Robert C. Martin'),
(3, 'Design Patterns', 'Erich Gamma');
UPDATE LibraryBook SET book_title = 'Clean Architecture' WHERE book_id = 2;
DELETE FROM LibraryBook WHERE book_id = 3;
SELECT * FROM LibraryBook;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| librarybook          |
+-----+

+-----+-----+-----+
| book_id | book_title          | author_name      |
+-----+-----+-----+
|      1 | Database System Concepts | Silberschatz    |
|      2 | Clean Architecture     | Robert C. Martin |
+-----+-----+-----+

```

7.Student club table:

```

CREATE TABLE StudentOrganization (
    club_id INT PRIMARY KEY,
    club_name VARCHAR(100) NOT NULL,
    budget DECIMAL(10, 2)
);
SHOW TABLES;
INSERT INTO StudentOrganization (club_id, club_name, budget) VALUES
(1, 'Robotics Club', 5000.00),
(2, 'Debate Society', 1500.50),
(3, 'Photography Club', 2000.00);
UPDATE StudentOrganization SET budget = 2500.00 WHERE club_id = 3;
DELETE FROM StudentOrganization WHERE club_id = 2;
SELECT * FROM StudentOrganization;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| studentorganization   |
+-----+
+-----+-----+-----+
| club_id | club_name      | budget |
+-----+-----+-----+
|      1 | Robotics Club  | 5000.00 |
|      3 | Photography Club | 2500.00 |
+-----+-----+-----+

```
